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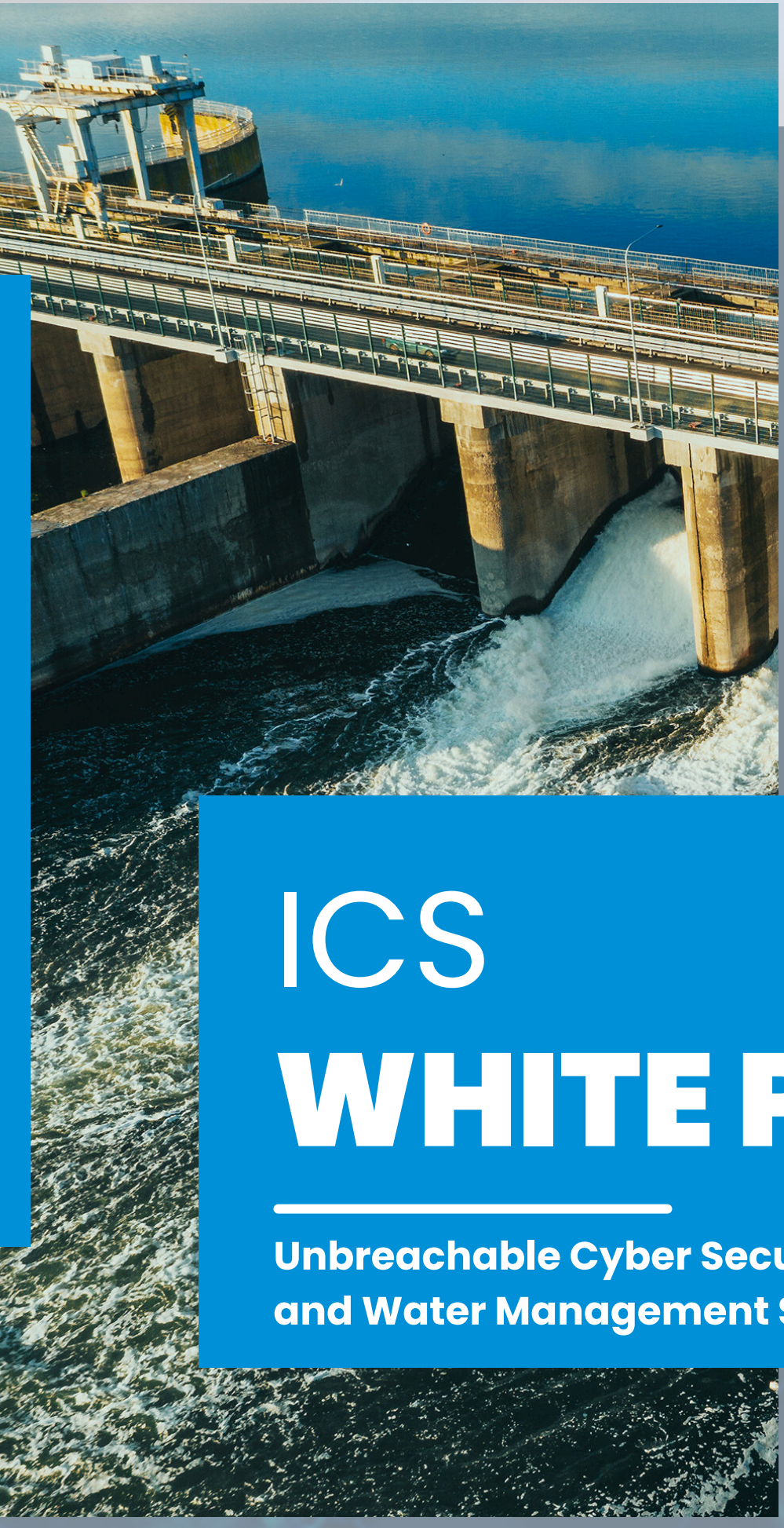
Dams



ICS
WHITE PAPER

Unbreachable Cyber Security for Hydropower and Water Management Systems

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COMPANY OVERVIEW

About Us

Terafence Private Limited is an Indian-owned entity engaged in R&D and manufacturing of FPGA-based Data Diodes through a technology transfer agreement with Terafence Limited, Israel.

It focuses on providing cybersecurity solutions for IoT and IIoT end-devices.

Mission

Terafence is committed to ensuring cybersecurity by enabling full control over data flow, offering near Air-Gap segmentation between networks at the hardware level. The goal is to maintain the functionality of end-devices while preventing unauthorized access.

Vision

Terafence envisions a secure digital landscape where IoT and IIoT devices are protected through isolation and segmentation at the lowest OSI layer. The company aims to eliminate cyber threats by removing vulnerable devices from attackers' reach without compromising their performance.

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ICS THREAT ANALYSIS

Supervisory Control and Data Acquisition (SCADA) systems and Programmable Logic Controllers (PLCs) are foundational to modern dam operations. A single compromised PLC—costing less than $200—can trigger cascading failures: opening floodgates, disrupting reservoir management, and causing infrastructure damage worth millions, along with severe risks to human safety.

Cyberattacks on water infrastructure, including dams, are no longer theoretical. If an attacker gains access to a dam's control network, they could alter water levels, disable alarm systems, or manipulate turbine operations—leading to flooding, power outages, or environmental damage. The consequences are not only financial but also reputational and life-threatening.

While external threats are widely discussed, studies consistently show that up to 80% of breaches originate internally. A seemingly innocent USB drive, a contractor's compromised laptop, or an engineer accessing public Wi-Fi may introduce malware into isolated control networks, bypassing perimeter defenses like firewalls and intrusion detection systems.

With the push toward smart infrastructure and remote monitoring, more dams are integrating IoT sensors, flow control automation, and cloud-based analytics. While this enhances performance and predictive maintenance, it also broadens the attack surface. Without secure data transmission, endpoint protection, and strict access controls, these improvements can become liabilities.

Historically, operational technology (OT) and information technology (IT) in dam systems were separate. However, today's data-driven decision-making demands tighter integration. This convergence, if done without a solid cybersecurity framework, increases exposure to threats and can delay digital transformation due to operational risk.

Real-world attacks on ICS systems around the globe demonstrate that the threat is no longer hypothetical. Attackers use phishing, remote code execution, and physical access to gain control over critical systems.

Dams—being high-impact, low-visibility targets—must now be defended with proactive strategies including network segmentation, zero-trust access, continuous monitoring, and staff training to prevent minor gaps from becoming large-scale failures.

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CRITICAL ANALYSIS IN DAM INFRASTRUCTURE

Introduction

India's dams are crucial for water management, agriculture, hydroelectric power, and flood control. As these infrastructures increasingly adopt automation and digital technologies like SCADA systems and PLCs, they become more vulnerable to cyber threats. Ensuring robust cybersecurity for these critical assets is vital for national security and public safety.

Digital Evolution and Its Risks

The adoption of real-time monitoring and automation in dam systems offers operational advantages but exposes them to various cybersecurity risks, such as:

- Unauthorized Access – Weak authentication or inadequate network segmentation could allow unauthorized control.
- Ransomware & Malware – Malicious software could disrupt operations, damage equipment, or cause operational paralysis.
- Insider Threats – Employees or contractors misusing access could intentionally or unintentionally compromise systems.
- Third-Party Vulnerabilities – Outdated software and unsecured third-party integrations can provide entry points for attackers.

India's Exposure to ICS Threats

While no major cyberattacks on dams have been reported in India, the rise in cyber incidents across critical infrastructure sectors points to increasing risks. The integration of cloud data and remote monitoring enhances efficiency but also amplifies the attack surface. Even a small breach, such as a compromised USB drive, could take down entire dam operations.

Key Security Recommendations

1. **Access Control & Authentication –** Enforce strict authentication protocols and multi-factor authentication to limit access.
2. **Secure Patch Distribution –** Implement secure distribution methods for patches to address vulnerabilities without compromising system integrity.
3. **Internal Awareness –** Regular training for staff on cybersecurity best practices and threat identification.
4. **Incident Response Plan –** Develop and test response protocols to contain and mitigate cyber incidents swiftly.
5. **Secure IT/OT Convergence –** Ensure the secure integration of IT and OT systems with encryption, air gaps, and firewalls.

Conclusion

As India modernizes its dam infrastructure, cybersecurity must be prioritized. A breach could have catastrophic consequences, affecting public safety and national assets. To mitigate this risk, a comprehensive, secure cybersecurity framework tailored to dam systems is essential, ensuring the integrity and safety of critical infrastructure.

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DAM CYBERSECURITY ARCHITECTURE

A hydroelectric authority operates **05 geographically distributed** dam sites. Each site contains **PLC** and **SCADA** systems to control floodgates, water levels, turbines, and other critical operations.

To maintain operational oversight and meet compliance standards, logs and sensor data must be transferred securely to a central analysis and monitoring center, without risking exposure to malware or remote threats.

Step-by-Step Architecture Flow

1. At Each Dam Site (OT Network):

- **PLC** and **SCADA** devices generate logs (e.g., gate position, pressure, flow rate).
- These logs are sent to the ***OT DMZ***, where:
  - The ***OT Historian*** aggregates and normalizes data.
  - The ***OT Firewall*** enforces strict access control.
  - The ***OT Security*** module monitors internal anomalies.

2. One-Way Transfer Using Terafence Data Diode:

- Logs are pushed unidirectionally via optical fiber from the ***OT DMZ*** to the ***IT DMZ*** using the **Terafence Data Diode**.
- Our diode physically enforces one-way communication, preventing any command or malware from reaching the ***OT systems***.

3. In the IT DMZ:

- A ***SIEM/Syslog*** server collects the logs.
- The ***IT Firewall*** applies access policies to regulate incoming log traffic.
- ***IT Security*** tools analyze and flag any abnormalities (e.g., unexpected gate movement, anomalous pressure spikes).

4. In the Corporate Network:

- Logs are forwarded to ***Central Management*** for high-level oversight.
- Analysts use workstations for performance tracking, risk detection, and policy updates.
- Data may optionally be pushed to the ***Internet (Cloud)*** for dashboards or reports—without any exposure to the ***OT systems.***

Key Cybersecurity Features of This Setup:

- Air-gapped security with hardware-enforced unidirectional data diode.
- No malware, patching, or remote access risk to dam control systems.
- Centralized visibility without compromising site isolation.

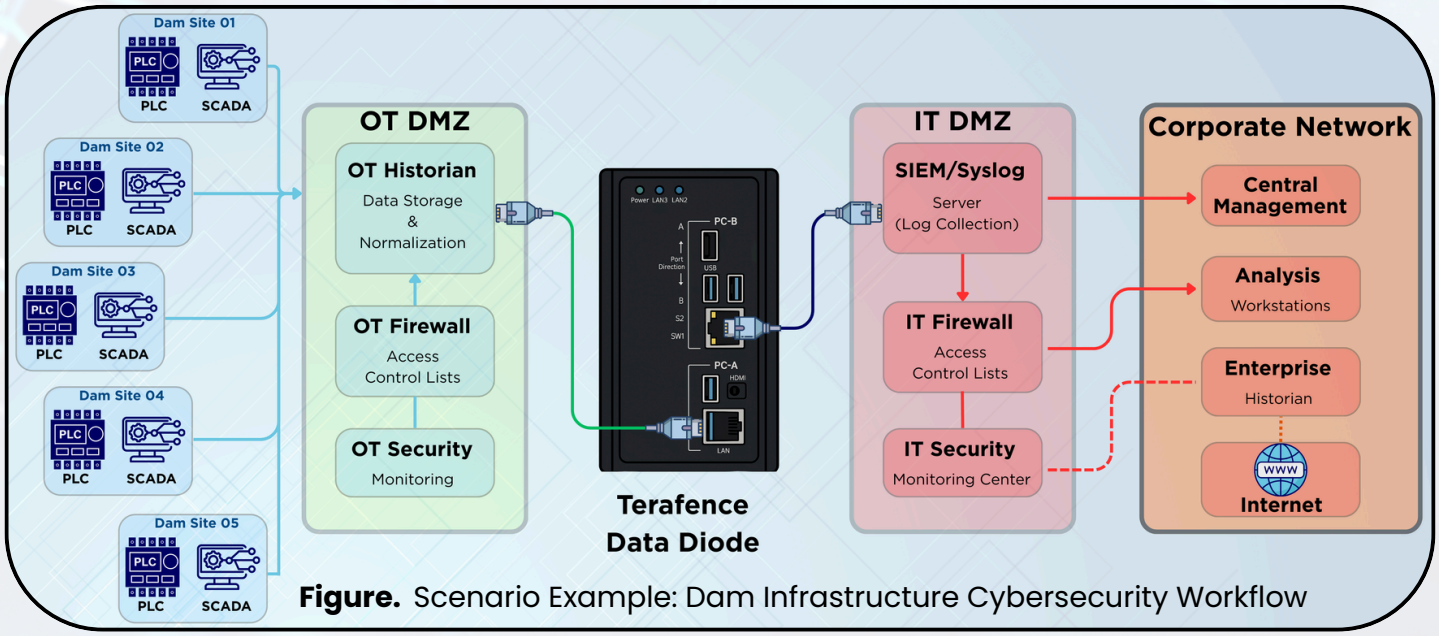

Figure. Scenario Example: Dam Infrastructure Cybersecurity Workflow

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SYSLOG CASE STUDY IN DAM INFRASTRUCTURE

As India's dam infrastructure continues to adopt advanced automation systems like SCADA and PLCs, the risk of cyber threats—such as unauthorized access and remote attacks—grows significantly. These systems, essential for managing water flow, power generation, and overall operations, are increasingly vulnerable to external cyber risks. To mitigate these threats, data diodes provide an essential security solution by ensuring one-way data flow, which effectively isolates critical systems from potential cyberattacks. By preventing any inbound data traffic while allowing necessary outbound communication, data diodes significantly reduce the attack surface of sensitive infrastructure.

For example, **Dam A** deployed data diodes to create a secure barrier between its internal operational systems and the external internet. This one-way communication system ensures that internal systems controlling crucial operations—such as floodgate management and hydropower systems—are completely isolated from external networks, preventing unauthorized access or cyberattack attempts.

While traditional cybersecurity measures like malware detection and firewalls are essential for broader protection, data diodes provide a fundamental layer of defense by physically blocking any inbound data flow, which is a crucial line of defense against remote exploitation. This solution enhances the security posture of the dam's infrastructure without relying on complex or resource-intensive malware detection systems, making it ideal for securing critical environments where maintaining the integrity and safety of operations is paramount.

Figure. Uni-directional Solution

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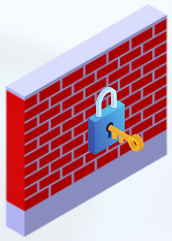
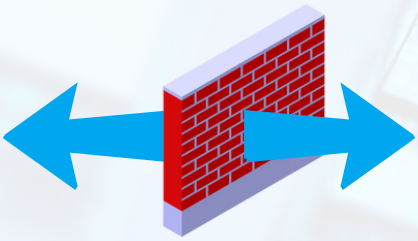
SECURITY– FIREWALL VS. TERAFENCE

"I can do this with a Firewall..."



Yes, you can configure a Firewall to deny any access from the outside, but:

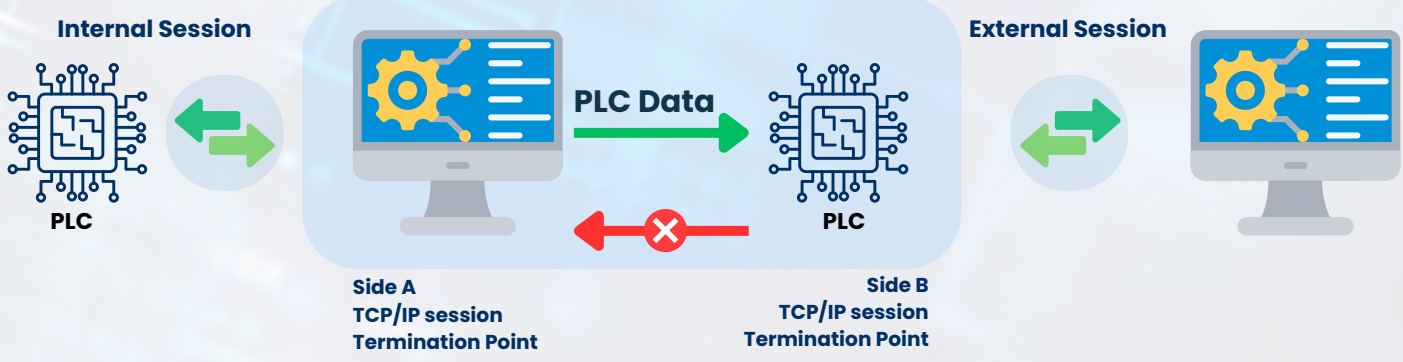
A Firewall is primarily a system that provides network security by filtering incoming and outgoing traffic based on user-defined rules. Its goal is to prevent unwanted communications while allowing legitimate ones with the correct key.



However, a Firewall will eventually grant access to an entity that meets its rule configuration. The problem is that once access is granted, the entity gains full access to the end-device, rendering the Firewall ineffective—unless it's a Layer 7 firewall. Essentially, the Firewall allows a live session between the end-device and an external entity, which is exactly what a hacker needs to manipulate the device.

Preventing all inbound traffic would turn the Firewall into a "unidirectional" device, but this disrupts protocol integrity between the IIoT device and the HMI. These must communicate to function properly. For example, in ModBus, the HMI sends a POLL/READ command to request parameters, and the PLC responds. Without a mediator, this process fails, and no Firewall can mediate such protocols.

With Terafence, no entity ever gains full access to the end-device. Terafence actively acts as a mediator between the PLC and HMI, unlike any Firewall. In special applications, it may allow only heavily filtered commands, which are accepted, filtered, disassembled, and securely forwarded via an out-of-band channel. The actual TCP/IP session terminates on the Terafence side next to the sender (External Session), ensuring no live sessions exist, thus preventing manipulation.



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